

Example 5: Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $a_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$ and $b = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$. Determine

whether b can be written as a linear combination of a_1 and a_2 . That is, determine whether weights x_1 and x_2 exist such that

$x_1 a_1 + x_2 a_2 = b$ (1) If vector equation (1) has a solution, find it.

Solution: Use the definitions of scalar multiplication and vector addition to rewrite the vector equation

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

$\uparrow$ $\uparrow$ $\uparrow$
 a_1 a_2 b

which is the same as

$$\begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

and

$$\begin{bmatrix} x_1 + 2x_2 \\ -2x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \quad (2)$$

The vectors on the left and right side of (2) are equal if and only if their corresponding entries are both equal. That is, x_1 and x_2 make the vector equation (1) true if and only if x_1 and x_2 satisfy the system

$$\begin{cases} x_1 + 2x_2 = 7 \\ -2x_1 + 5x_2 = 4 \\ -5x_1 + 6x_2 = -3 \end{cases} \quad (3)$$

To solve this system, row reduce the augmented matrix of the system

$$\left[\begin{array}{ccc|c} 1 & 2 & 7 & 7 \\ -2 & 5 & 4 & 4 \\ -5 & 6 & -3 & -3 \end{array} \right] \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 2 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} \\ + 5 \times \text{Row 1}} \left[\begin{array}{ccc|c} 1 & 2 & 7 & 7 \\ 0 & 9 & 18 & 18 \\ 0 & 16 & 32 & 32 \end{array} \right]$$